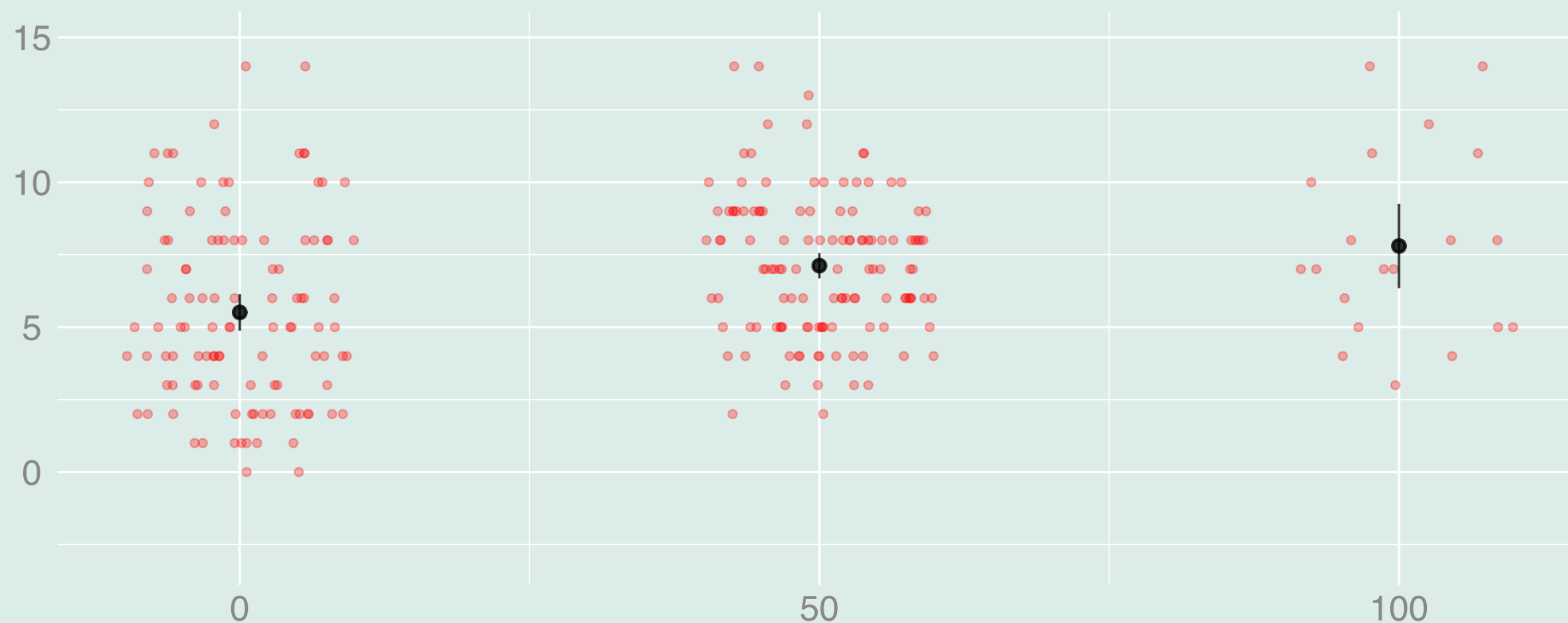


QTM 220: Lecture 13

Misspecification and Statistical Inference

Review

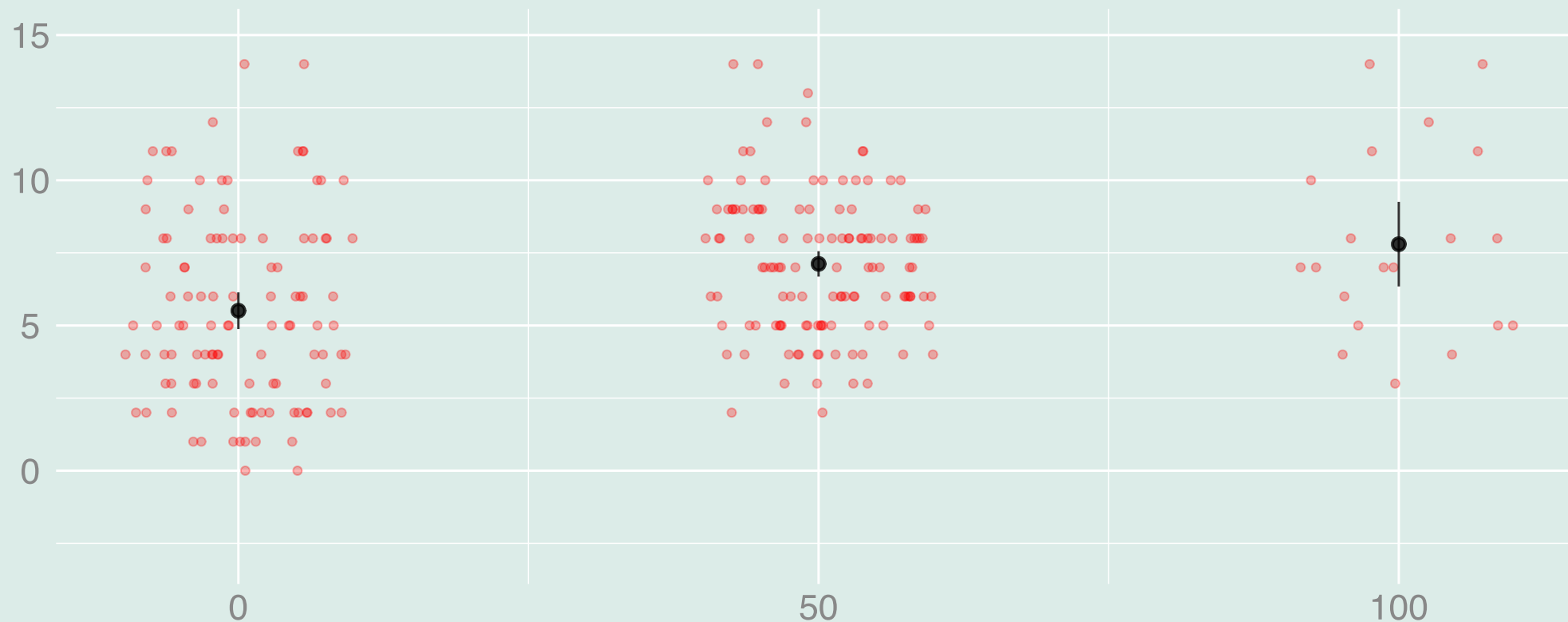
Estimating An Average Treatment Effect



- We reduced the problem to estimating a difference in subpopulation means.
- And we tried estimating that using a difference in subsample means.
- I've drawn these, $\pm$ two standard errors, in black.

$$\tau(100) = \frac{1}{m_{100}} \sum_{j:d_j=100} y_j - \frac{1}{m_{50}} \sum_{j:d_j=50} y_j \approx \hat{\tau}(100) = \frac{1}{N_{100}} \sum_{i:D_i=100} Y_i - \frac{1}{N_{50}} \sum_{i:D_i=50} Y_i$$

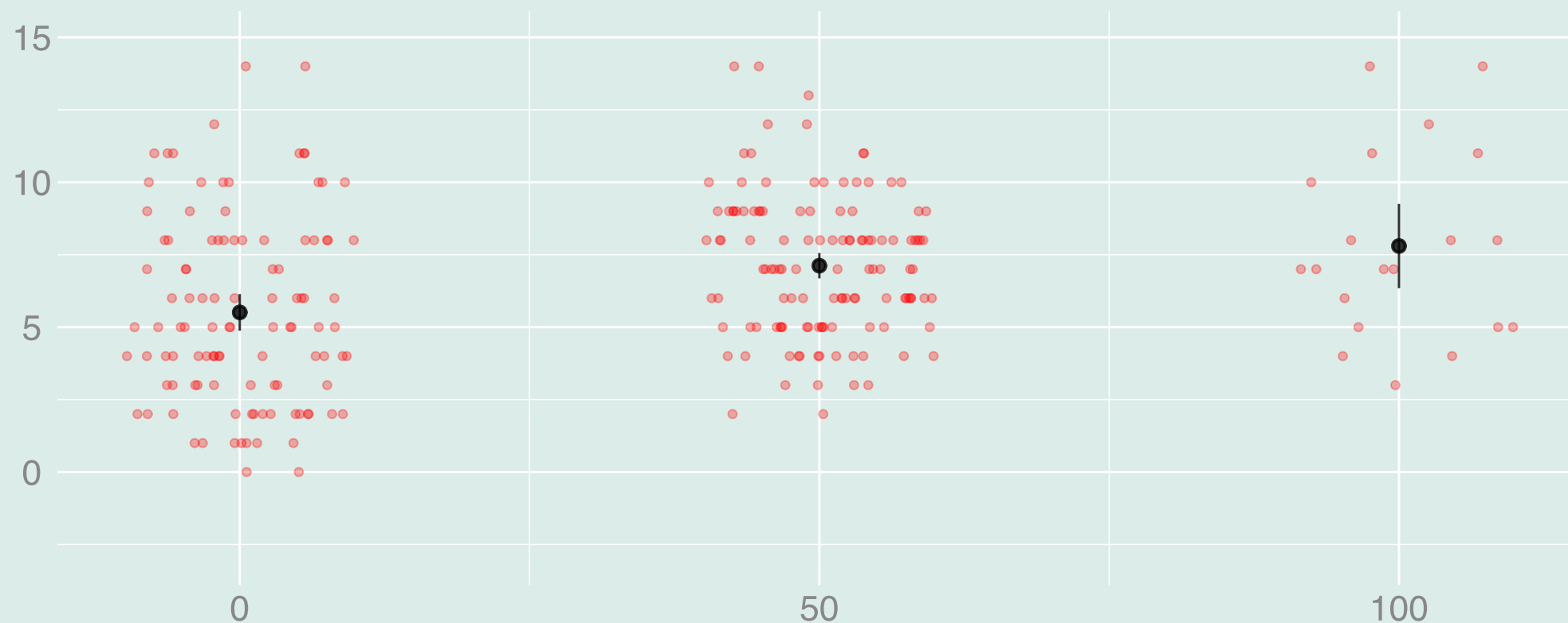
Dissapointing Results



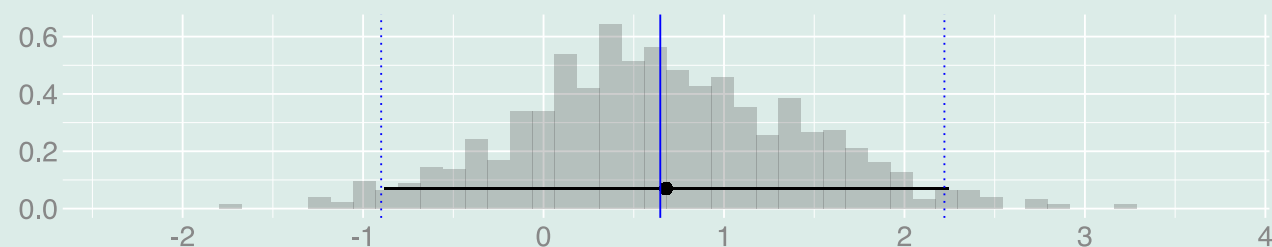
- We weren't happy with the resulting precision.
 - The problem was that we had too few observations with the expensive treatment ($D_i = 100$).
 - This made the subsample mean a high-variance estimate of the corresponding subpopulation mean.

$$V[\hat{\tau}(100)] = V[\hat{\mu}(50)] + V[\hat{\mu}(100)] \geq V[\hat{\mu}(100)] = \frac{V[Y \mid D_i = 100]}{N_{100}} \approx 0.53 \approx 0.73^2$$

Uninformative Interval Estimates



- Our confidence intervals, whether calibrated via bootstrap or normal approximation, are wide.
 - One thing people find notable is that they contain both positive and negative numbers.
 - This means we can't say, with confidence, whether our treatment effect is positive or negative.

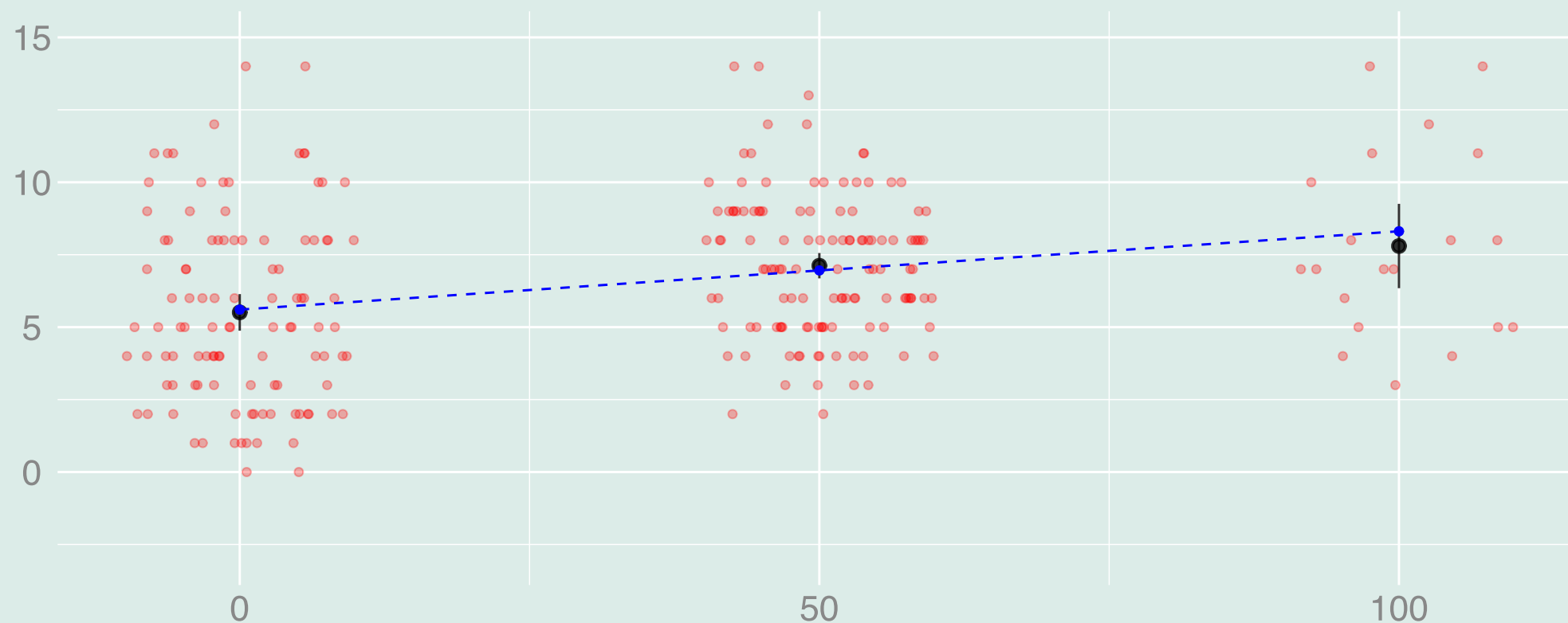


Bootstrap Sampling Distribution for the Difference-in-Means



Normal Approximation to the Sampling Distribution for the Difference-in-Means

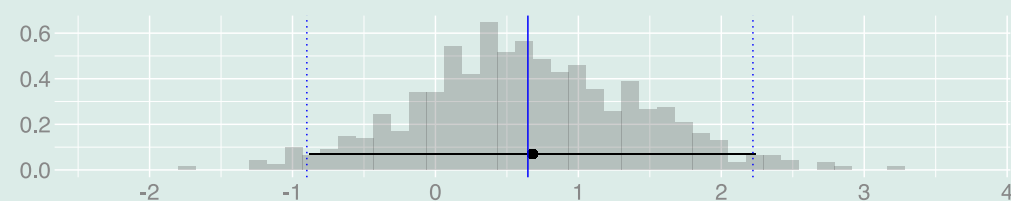
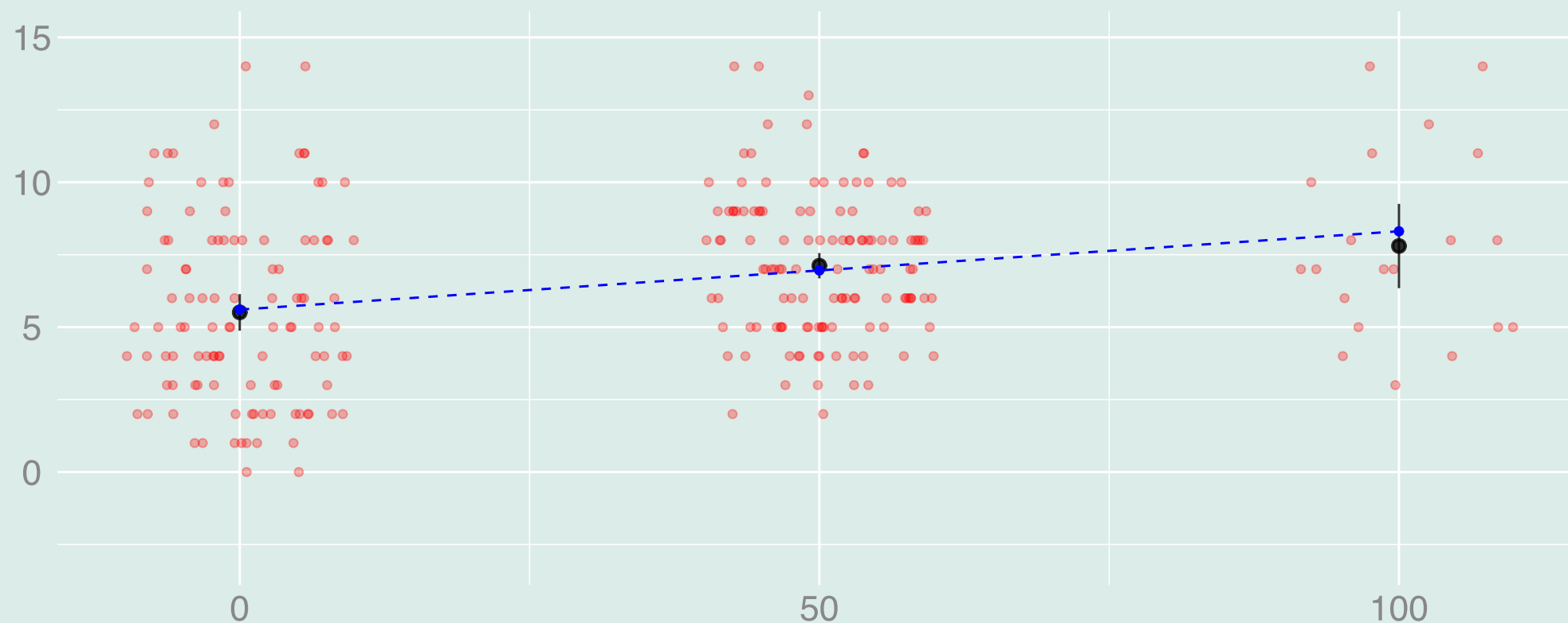
An Attempt At Being Clever



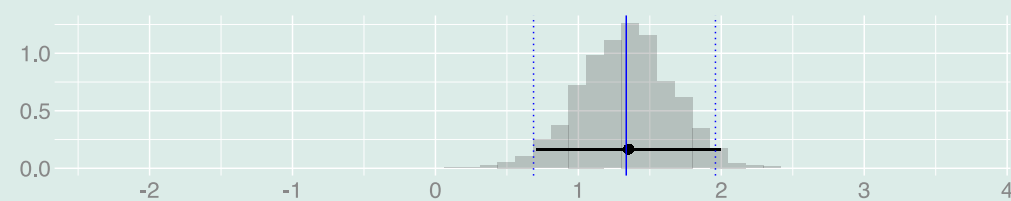
- To work around this, we tried estimating our subpopulation means using the [least-squares line](#).

$$\hat{\tau}(100) = \hat{\mu}(100) - \hat{\mu}(50) \quad \text{where} \quad \hat{\mu}(x) = \hat{a}x + \hat{b} \quad \text{for} \quad \hat{a}, \hat{b} = \operatorname{argmin}_{a,b} \sum_{i=1}^n (aX_i + b - Y_i)^2$$

An Attempt At Being Clever



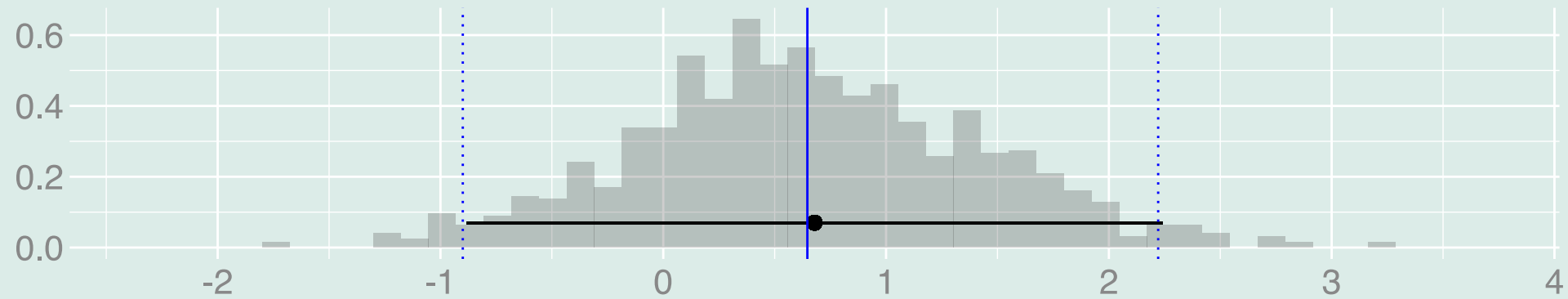
Difference in Means Bootstrap Sampling Distribution



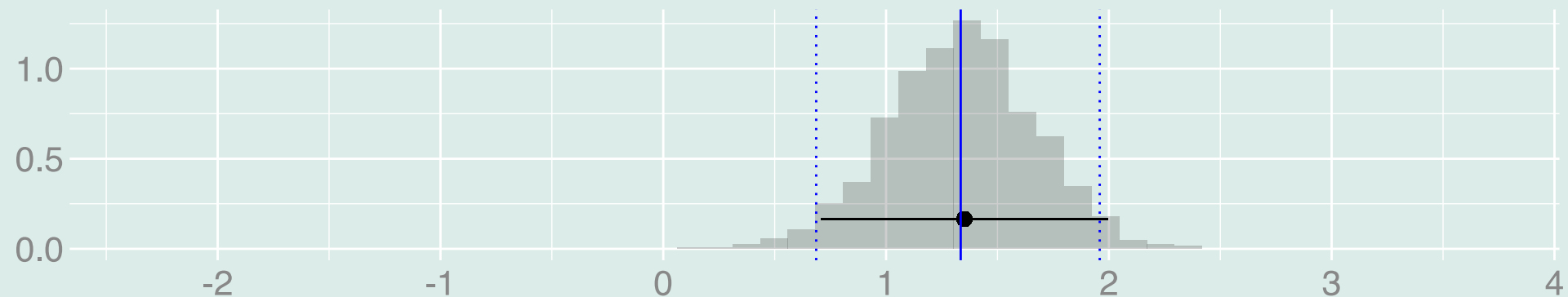
Fitted-Line-Based Bootstrap Sampling Distribution

- This *did* reduce variance.
 - There are lots of observations for cheaper treatments.
 - These helped us get a more precise estimate of the line's slope.
- But the line gave us pretty different estimates of the subpopulation means.
 - We were higher at $D_i = 100$ and low at $D_i = 50$.
 - So the difference, high - low, was bigger.

Inconsistent Interval Estimates



Difference in Means Bootstrap Sampling Distribution



Fitted-Line-Based Bootstrap Sampling Distribution

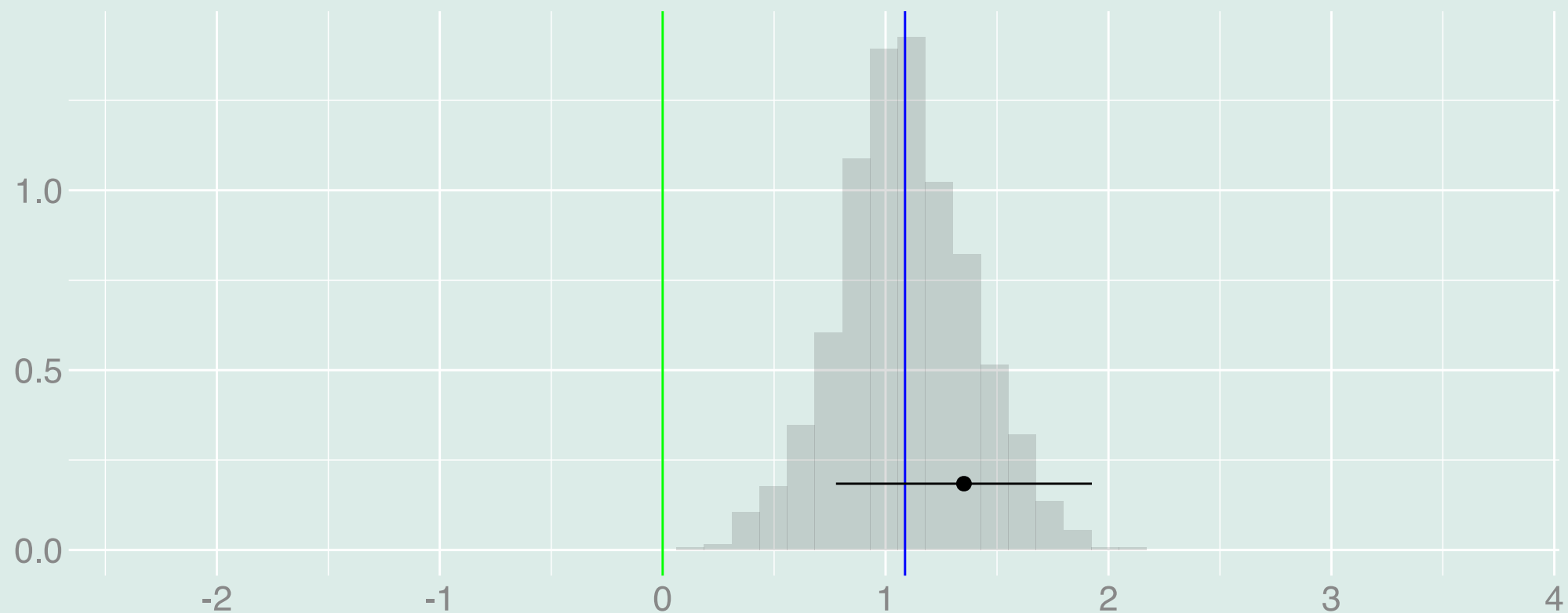
- These estimates are so different that our confidence intervals lead to meaningfully different claims.
- In particular, they say different things about whether our treatment effect is positive.
- The difference in subsample means approach would say we don't know.
 - A bootstrap-calibrated confidence interval based on that estimate covers zero.
- The fitted line approach would say we do know that it's positive.
 - A bootstrap-calibrated confidence interval based on that approach is to the right of zero.

That's Wrong

- This is fake data, so we can see that by looking at the code generating it.
- In the data we're looking at ...
 - The actual effect is exactly zero.
 - The potential outcomes $Y_i(50)$ and $Y_i(100)$ are *exactly the same*.

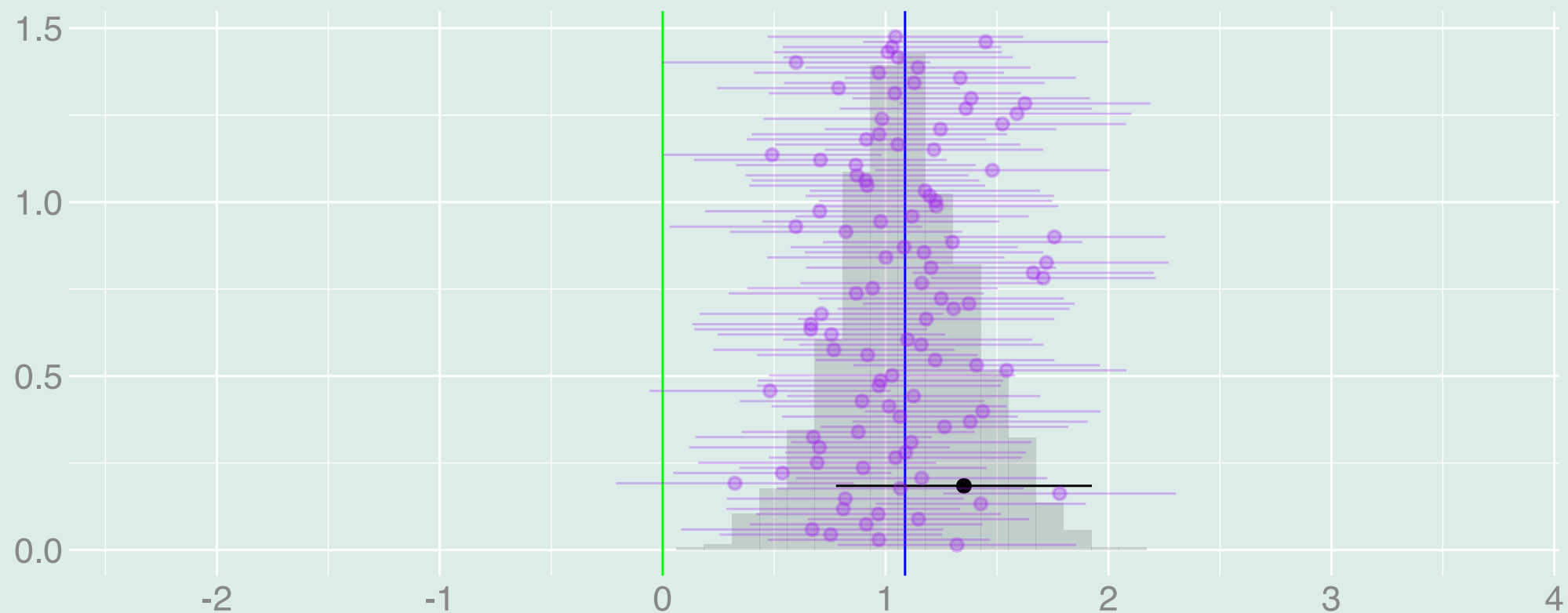
```
1 # Sampling 250 employees. Our population is an even mix of ...
2 #   'blue guys', who go 1/4 days no matter what
3 #   'green guys', who go more if it's free but --- unlike the ones in our table earlier --- don't care about classes
4
5 n=250
6 is.blue = rbinom(n,1, 1/2)
7 Yblue   = rbinom(n, 30, 1/4)
8 Ygreen0 = rbinom(n, 30, 1/10)
9 Ygreen50 = rbinom(n, 30, 1/5)
10
11 Y0      = ifelse(is.blue, Yblue, Ygreen0)
12 Y50     = ifelse(is.blue, Yblue, Ygreen50)
13 Y100    = Y50    ### <--- this tells us we've got no effect.
14
15 # Randomizing by rolling two dice. R is their sum
16 R = sample(c(1:6), size=n, replace=TRUE) + sample(c(1:6), size=n, replace=TRUE)
17 D = case_match(R, 1:6~0, 7:10~50, 11:12~100)
18 Y = case_match(D, 0~Y0, 50~Y50, 100~Y100)
```

The Line-Based Estimator is **Biased**



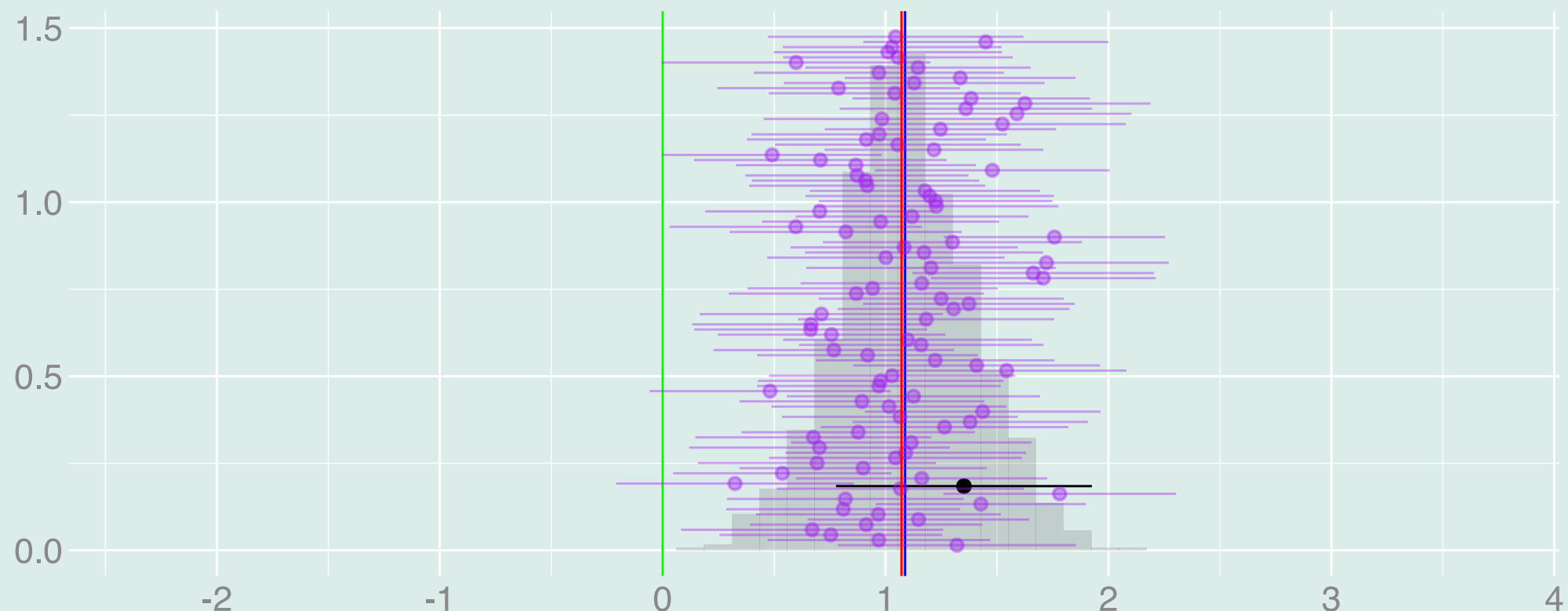
- We can do a little simulation to see that.
- We'll generate 1000 fake datasets like the one we've been looking at.
 - And calculate our estimator for each one.
 - This gives us draws from our estimator's *actual* sampling distribution.
 - I've used them to draw a histogram.
- The mean of these point estimates is **1.09**.
 - Since our actual effect is zero, that's our bias. The blue line.
- Our actual point estimate is off in the same direction—but a bit more than usual.
 - It's shown in black with a corresponding 95% interval.
 - An interval that fails to cover the actual effect of zero. The green line.

The Resulting Coverage Is **Bad**



- We can construct 95% confidence intervals the same way in each fake dataset.
 - I've drawn 100 of these intervals in purple.
 - You can see that they sometimes cover zero, but not as often as you'd like.
- $30 / 1000 \approx 3\%$ of them cover zero. That's not good.
- If that's typical of our analysis, people really shouldn't trust us.
- So why have statisticians gotten away with fitting lines for so long?
- In essence, it's by pretending we were trying to do something else.

What We Pretend To Do



- Our estimator — 50 times the slope of the least squares line — *is* a good estimator of something.
- It's a good estimate of the analogous population summary: 50 × the *population least squares slope*.
- That's the **red line**. 924 / 1000 ≈ 92% of our intervals cover it. That sounds pretty good.

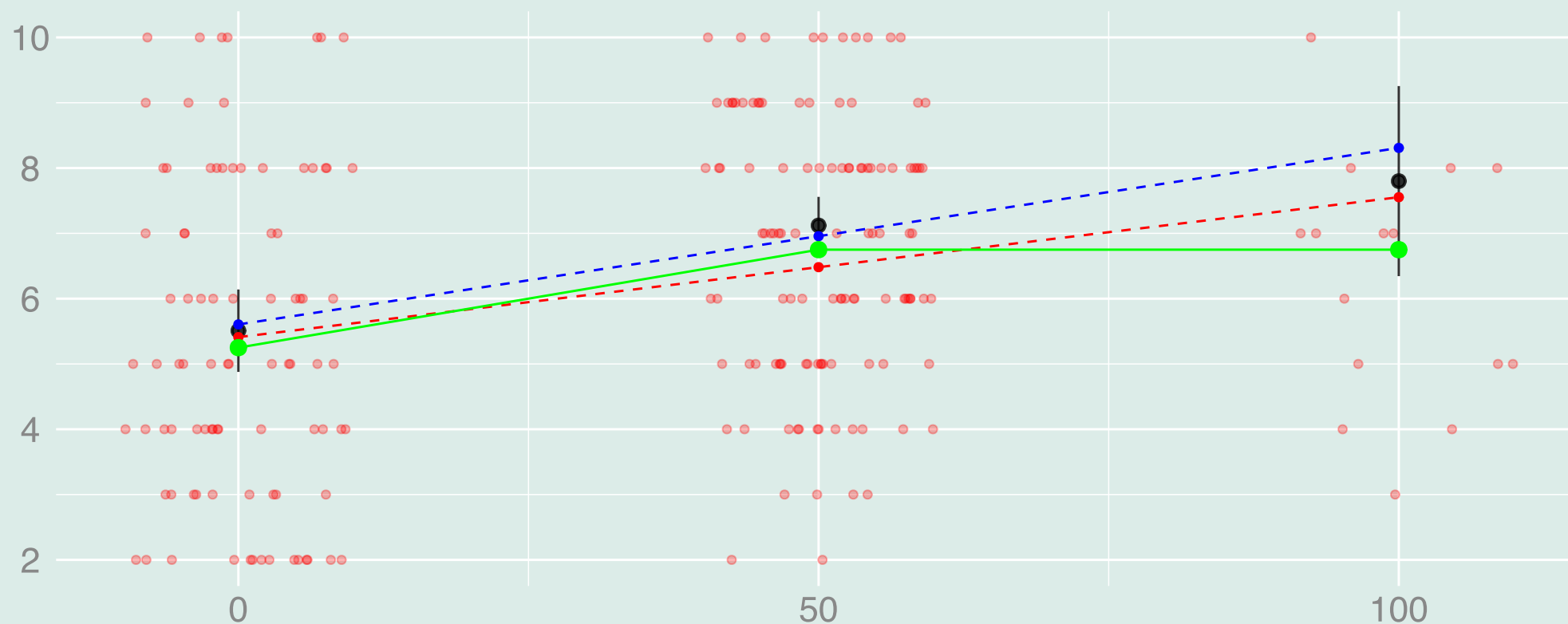
$$\hat{\tau}(100) = \hat{\mu}(100) - \hat{\mu}(50) = (100\hat{a} + \hat{b}) - (50\hat{a} + \hat{b}) = 50\hat{a}$$

$$\text{where } \hat{\mu}(x) = \hat{a}x + \hat{b} \text{ for } \hat{a}, \hat{b} = \operatorname{argmin}_{a,b} \sum_{i=1}^n (aX_i + b - Y_i)^2$$

$$\tilde{\tau}(100) = \tilde{\mu}(100) - \tilde{\mu}(50) = (100\tilde{a} + \tilde{b}) - (50\tilde{a} + \tilde{b}) = 50\tilde{a}$$

$$\text{where } \tilde{\mu}(x) = \tilde{a}x + \tilde{b} \text{ for } \tilde{a}, \tilde{b} = \operatorname{argmin}_{a,b} \mathbb{E}[(aX_i + b - Y)^2]$$

Problem: That's Not What We Wanted To Estimate

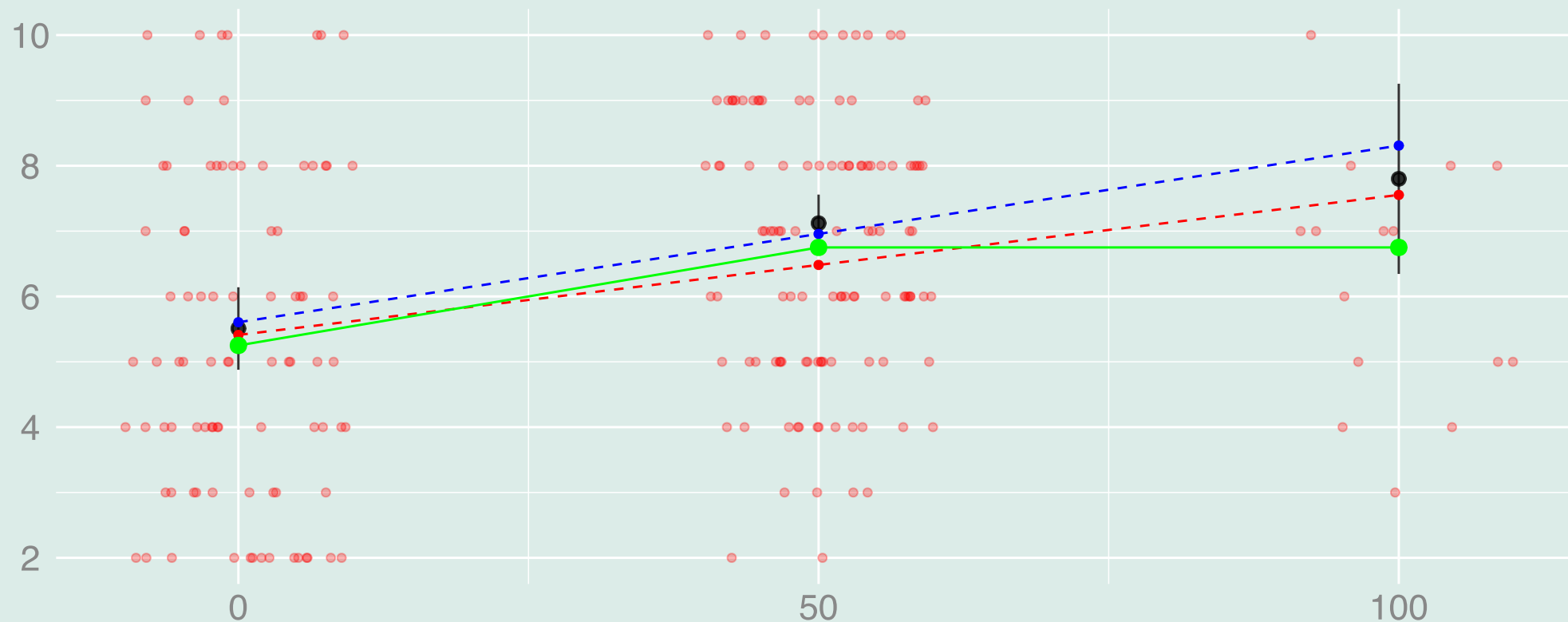


- The effects we want to estimate are, as a result of randomization, differences in subpopulation means.

$$\tau(50) = \mu(50) - \mu(0) \quad \text{and} \quad \tau(100) = \mu(100) - \mu(50)$$

- The population **least squares line** *doesn't go through these means*. It can't.
- They don't lie along a line. They lie along a **hockey-stick shaped curve**.
- This means no line—including the population least squares line—can tell us what we want to know.
- We call this **misspecification**. We've specified a shape that doesn't match the data.
- This problem has a history of getting buried in technically correct but opaque language.

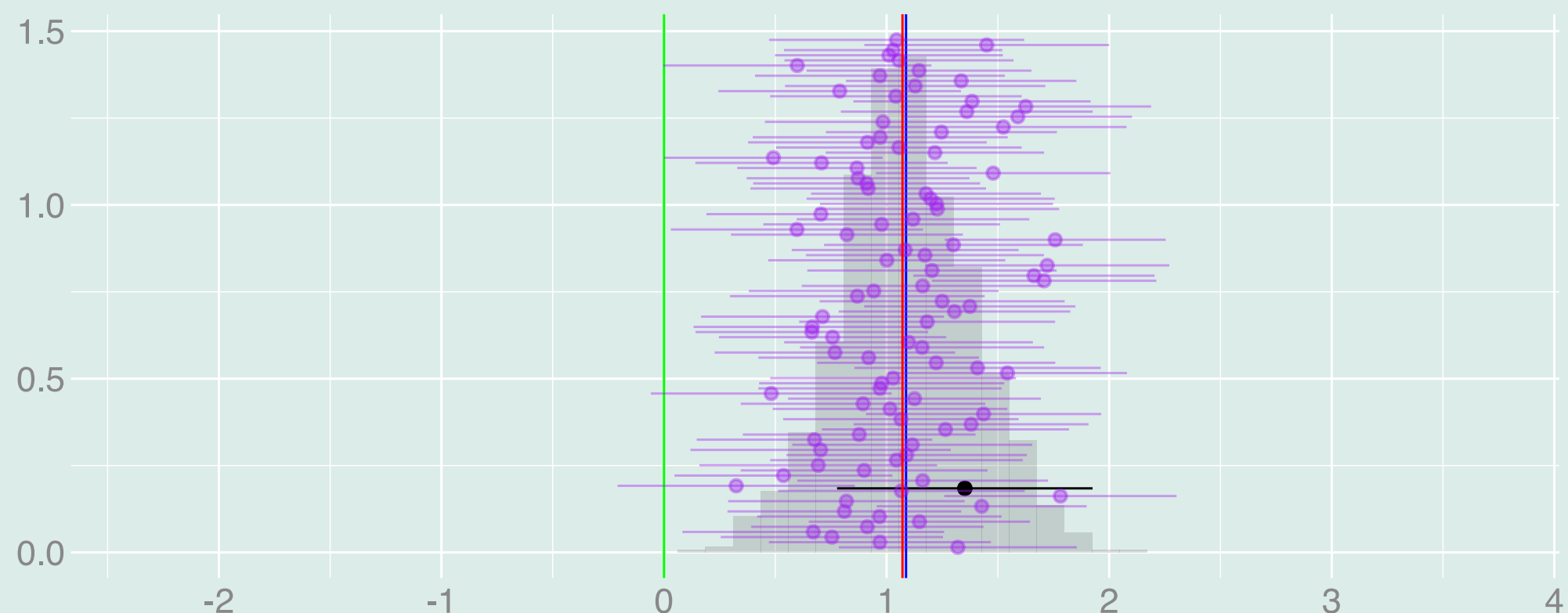
Language Games



- The widespread use of causal language is a recent development.
- In the past, people would do a little linguistic dance to avoid talking about causality.
- They'd talk about what they were really estimating as if it were what you wanted to know.
- This not only buried issues of causality, it buried issues of bias due to *misspecification*.
 - e.g., due to trying to use a line to estimate a hockey-stick shaped curve.
- They'd say 'The regression coefficient for Y on D is **0.03**.'
 - You'd be left to interpret this as saying the treatment effect was roughly 50×0.03 .
 - That's wrong, but it's 'your fault' for interpreting it that way. **They set you up.**

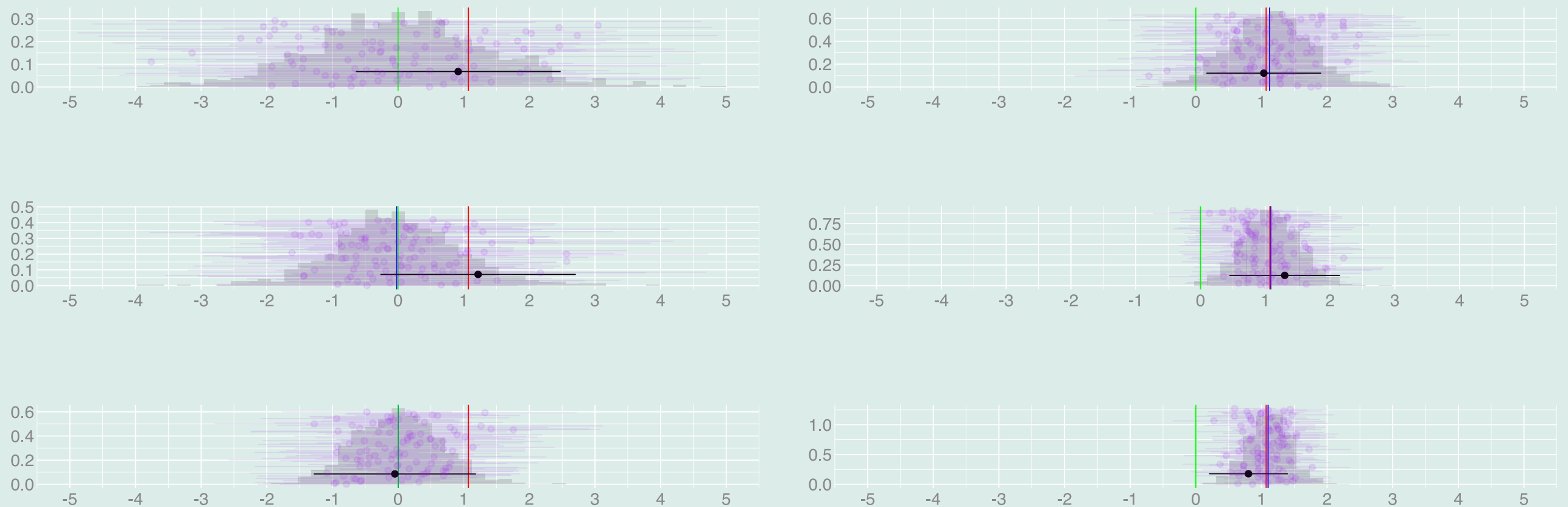
Today

We'll Think About Misspecification



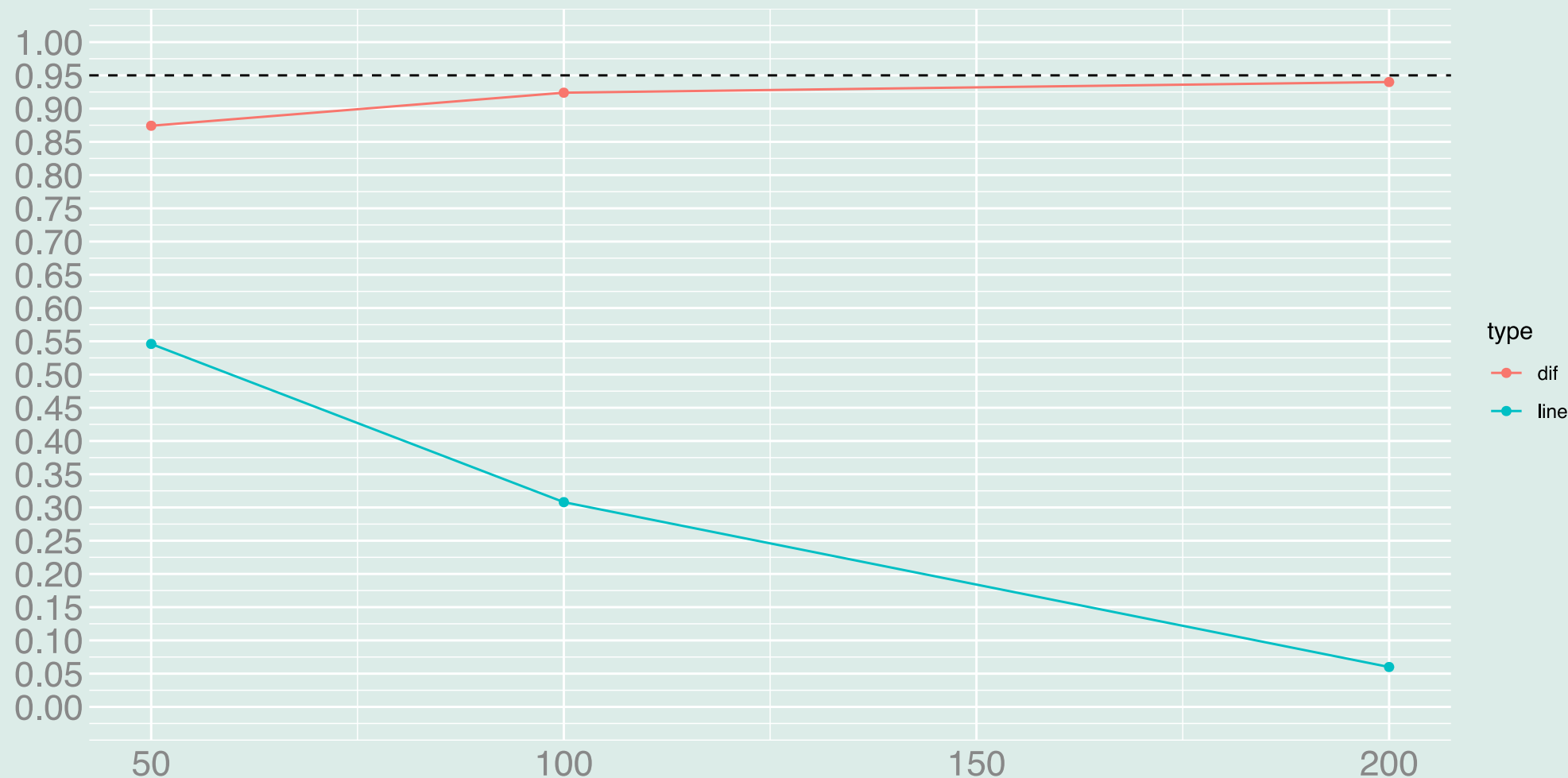
- In particular, we'll think about its implications for statistical inference. This boils down to...
- what happens when our estimator's sampling distribution isn't centered on what we want to estimate.
- We can see that, if our sampling distribution is narrow enough, the coverage of interval estimates is *awful*.
- Today, we'll think about how sample size impacts that. Something interesting happens.
- From a point-estimation perspective, more data is always good. Our estimators do get more accurate.
- But from an inferential perspective, it can be very bad.
- When we use misspecified models, our coverage claims will get *less accurate* as sample size grows.

The Impact of Misspecification as Sample Size Grows



- Here we're seeing the sampling distributions of our two estimators at three sample sizes.
 - Left/Right: Difference in Subsample Means / Difference in Values of Fitted Line
 - Top/Middle/Bottom: Sample size 50,100,200
- The green line is the actual effect.
- The red one is the effect estimate we get using the population least squares line: $50 \times$ the slope of the red line.
- I've drawn 100 confidence intervals based on normal approximation for each estimator.
- We can see that the line-based estimator's coverage gets increasingly bad as sample size increases.

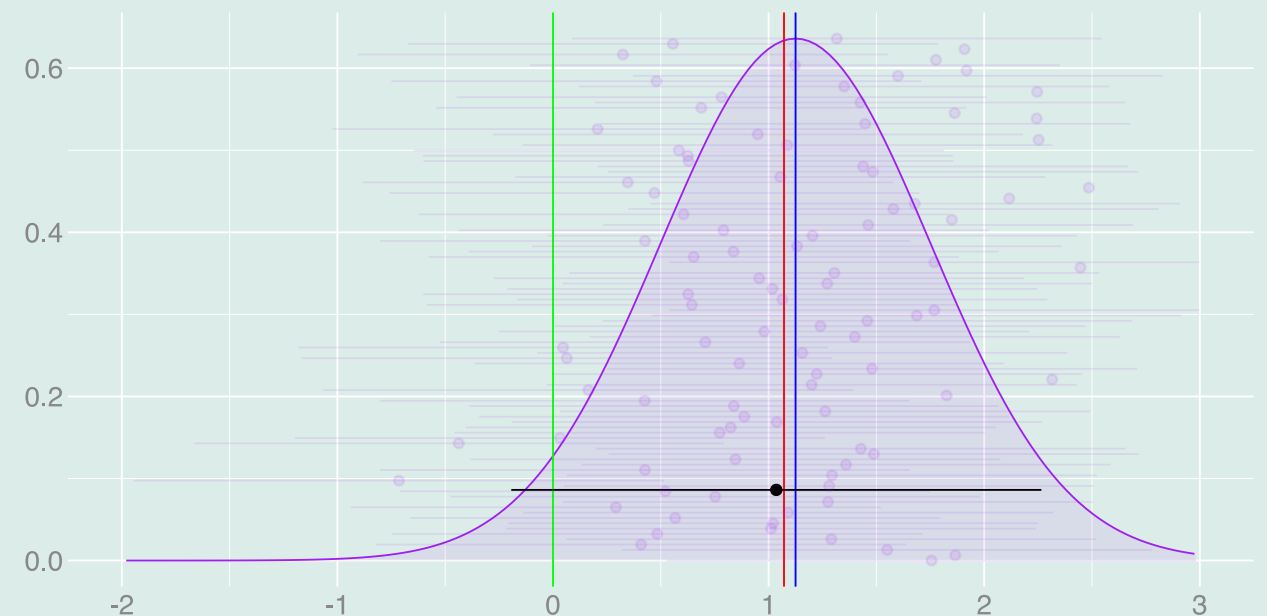
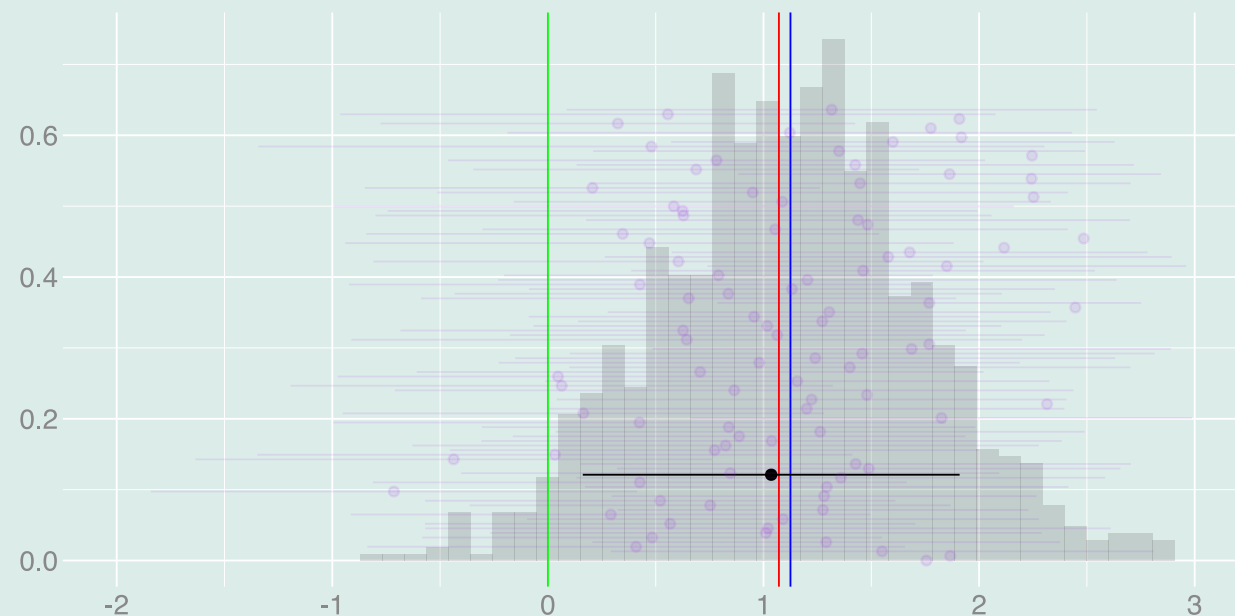
Coverage



- Here are the coverage rates for each estimator at each sample size.
- The difference in sample means works more or less as we'd like.
 - Coverage is always pretty good and is roughly 95% — what we claim — in larger samples.
- The line-based estimator does the opposite.
 - Coverage is bad in small samples and worse in larger ones.
- Now, we're going to do some exercises. We're going to work out how to calculate coverage by hand.
 - The point is to get some perspective on what it means to have 'just a little bit of bias'.
 - This'll come in handy when people around you say a bit of misspecification isn't a big deal.

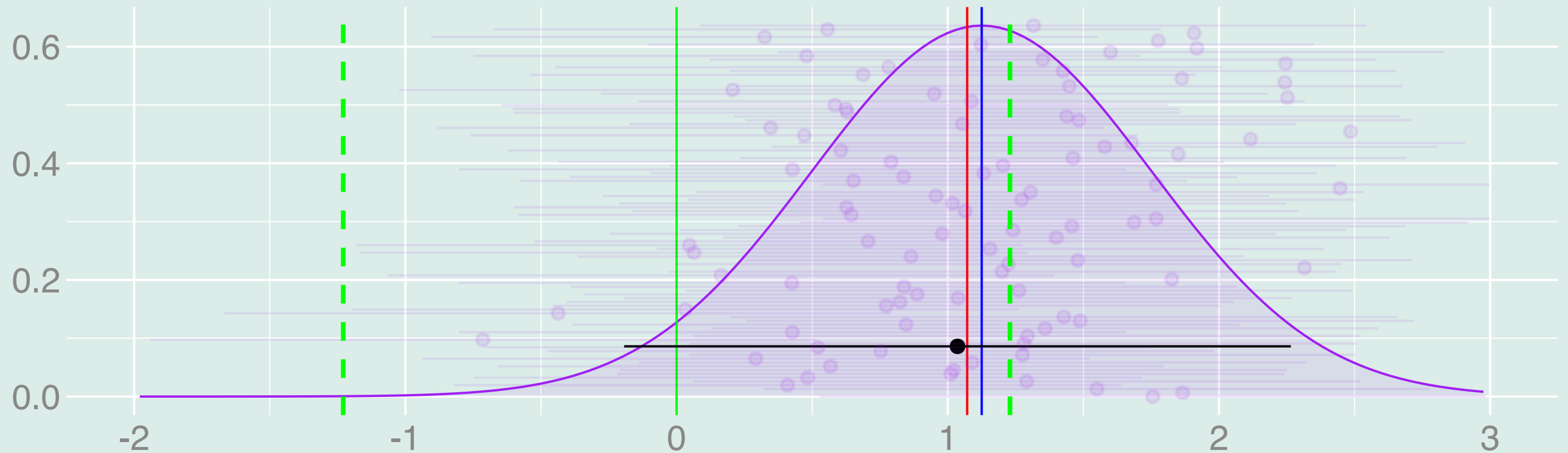
Exercises

Set Up



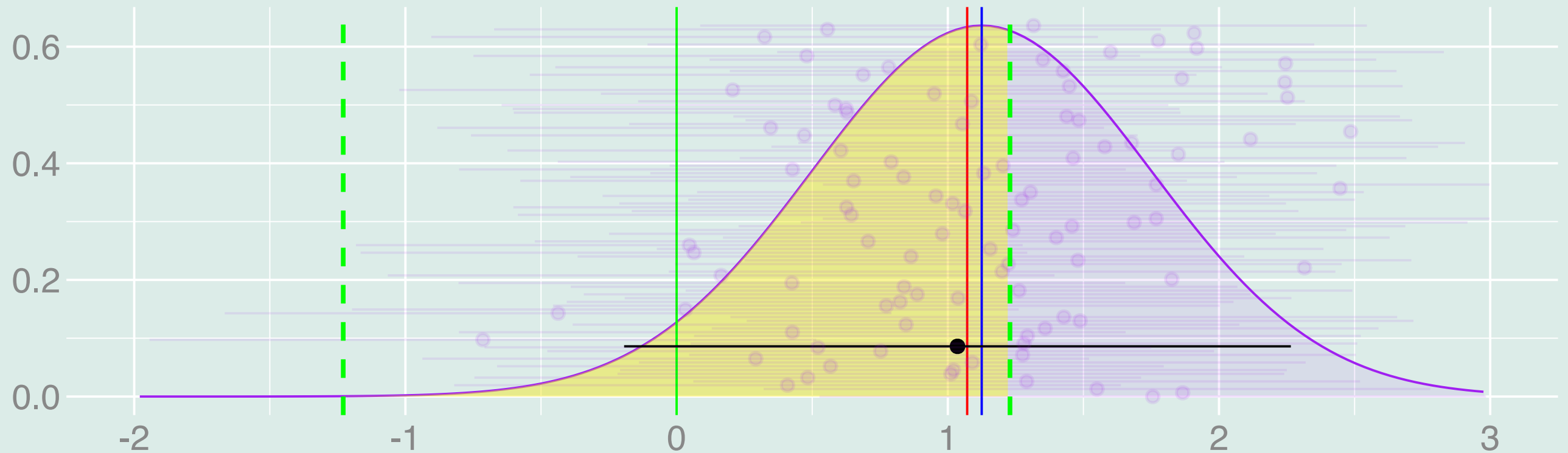
- Above, I've drawn the sampling distribution for the estimator $\hat{\tau}$ and a normal approximation to it.
- We're going to calculate that estimator's coverage rate. Approximately. We'll make some simplifications.
- We'll act as if the normal approximation to the sampling distribution were perfect.
- We'll assume our intervals are perfectly calibrated.
 - i.e., for a draw $\hat{\tau}$ from the sampling distribution ...
 - the interval is $\hat{\tau} \pm 1.96\sigma$ where σ is our estimator's actual standard error.

Exercise 1



- Let's think about things from the treatment effect's perspective. The solid green line.
- Let's give it arms the same length as our intervals have. They go out to the dashed green lines.
- An interval's arms touch the effect if and only if the effect touches the interval's center, i.e., the corresponding point estimate.
- So we're calculating the proportion of point estimates covered by *the treatment effect's arms*.
- **Exercise.** Write this in terms of the sampling distribution's probability density $f(x)$
 - This is a calculation you can illustrate with a sketch on top of the plot above. You may want to start there.

Exercise 1

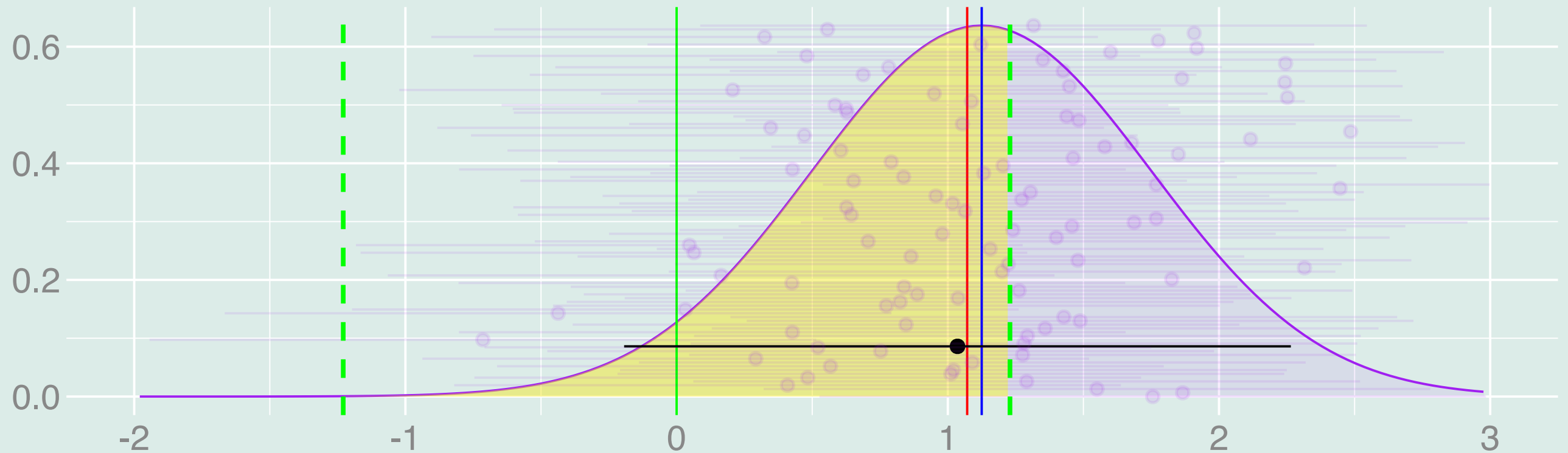


- To calculate the proportion of point estimates covered by *the treatment effect's arms*
- We integrate the sampling distribution's probability density from one to the other.

$$\text{coverage} = \int_{-1.96\sigma}^{1.96\sigma} f(x)dx \quad \text{where} \quad f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\tilde{\tau})^2}{2\sigma^2}}$$

- Here I've written in the actual formula of the density of the normal I drew.
 - The one with mean $\tilde{\tau}$ and standard deviation σ .
 - I'm not expecting you to have that memorized, but it does come in handy from time to time.
- This doesn't yet give us all that much *generalizable* intuition.
- But we'll derive some by, to some degree, simplifying that integral.

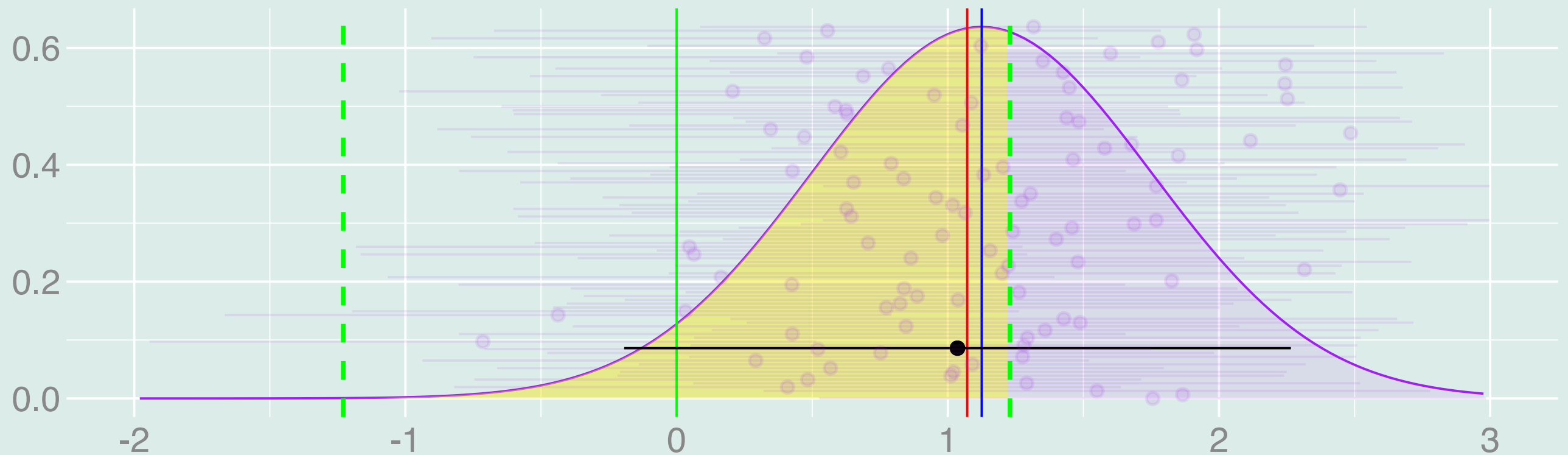
Exercise 2



- Now we'll rewrite our coverage as an integral of the [standard normal density](#)
- To do it, we're going back to intro calculus. We're making a *substitution* $u = (x - \tilde{\tau})/\sigma$.
- This means we have to change the limits.
 - As x goes from -1.96σ to 1.96σ ...
 - u goes from $-1.96 - \tilde{\tau}/\sigma$ to $1.96 - \tilde{\tau}/\sigma$.
- And make a substitution into the integrand.
 - For $(x - \tilde{\tau})^2/\sigma^2$, we substitute u^2 .
- And multiply by dx/du so we get to integrate over u .
 - $x = u\sigma + \tilde{\tau}$ so $dx = \sigma du$

$$\text{coverage} = \int_{-1.96\sigma}^{1.96\sigma} e^{-\frac{(x-\tilde{\tau})^2}{2\sigma^2}} dx$$

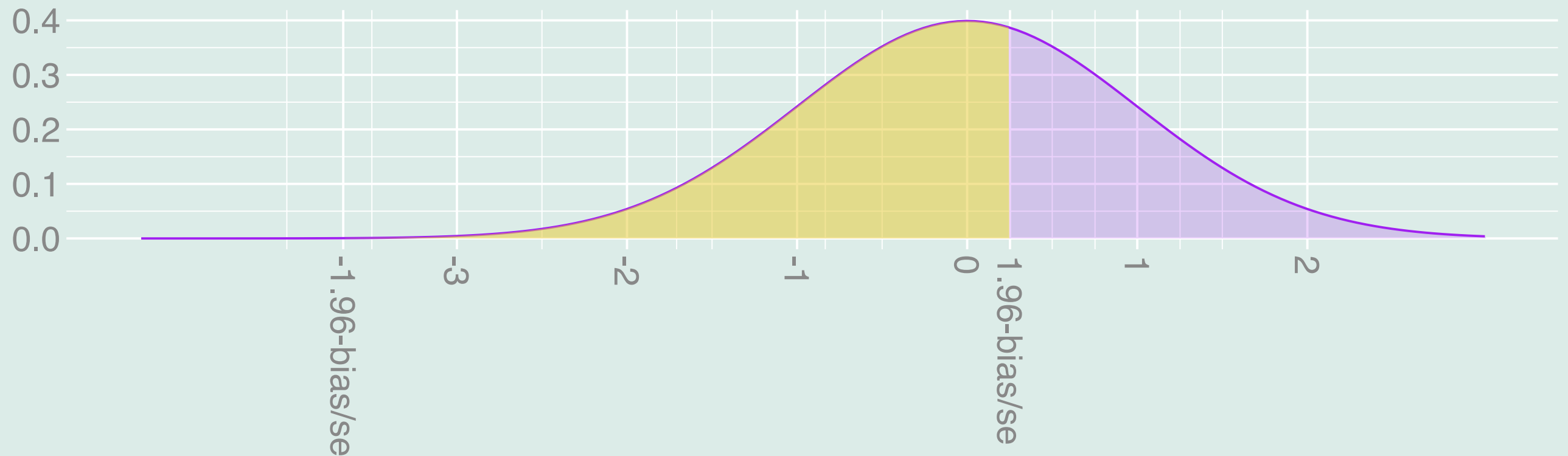
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$$\text{coverage} = \int_{-1.96\sigma}^{1.96\sigma} e^{-\frac{(x-\tilde{\tau})^2}{2\sigma^2}} dx = \int_{-1.96-\tilde{\tau}/\sigma}^{1.96-\tilde{\tau}/\sigma} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{u^2}{2}} \sigma du = \int_{-1.96-\tilde{\tau}/\sigma}^{1.96-\tilde{\tau}/\sigma} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$$

Summary and Generalization



- We can calculate the coverage of a biased, approximately normal estimator easily.
- It's an integral of the standard normal density over the range $\pm 1.96 - \text{bias}/\text{se}$.
- The **R** function call **pnorm(x)** calculates that integral from $-\infty$ to x . All we need to do is subtract.

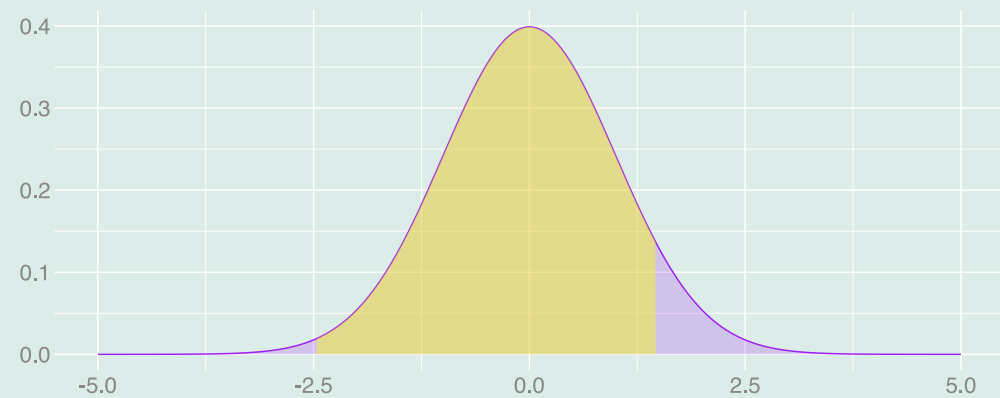
Math Version

$$\begin{aligned}\text{coverage} &= \int_{-1.96 - \frac{\text{bias}}{\text{se}}}^{+1.96 - \frac{\text{bias}}{\text{se}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du \\ &= \int_{-\infty}^{+1.96 - \frac{\text{bias}}{\text{se}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du - \int_{-\infty}^{-1.96 - \frac{\text{bias}}{\text{se}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du\end{aligned}$$

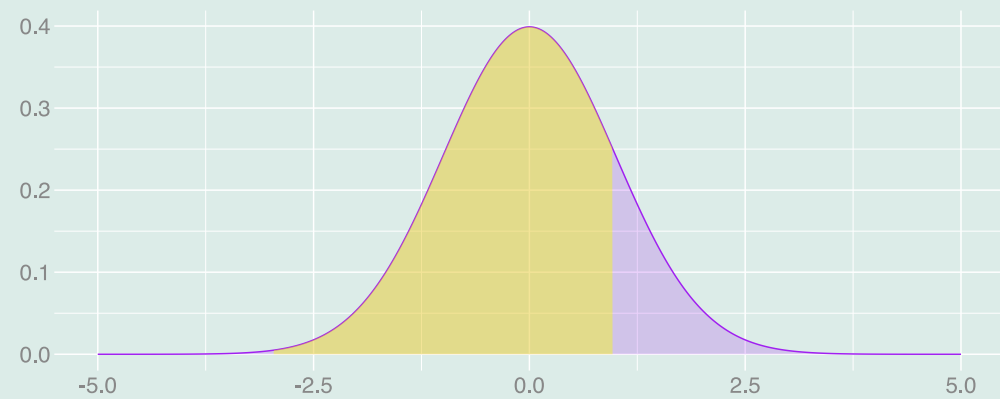
R Version

```
1 coverage = function(bias.over.se) { pnorm(1.96 - bias.over.se) - pnorm(-1.96 - bias.over.se) }
```

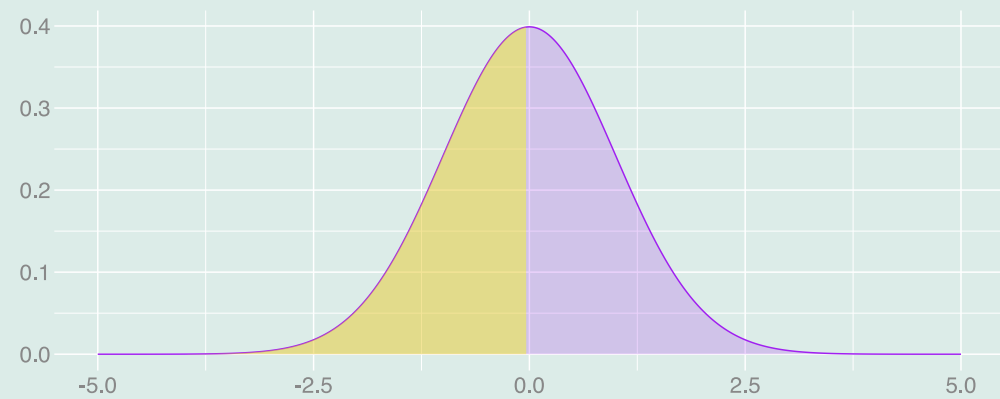
Examples



$$\frac{\text{bias}}{\text{se}} = \frac{1}{2} \implies \text{coverage} = 92\%$$

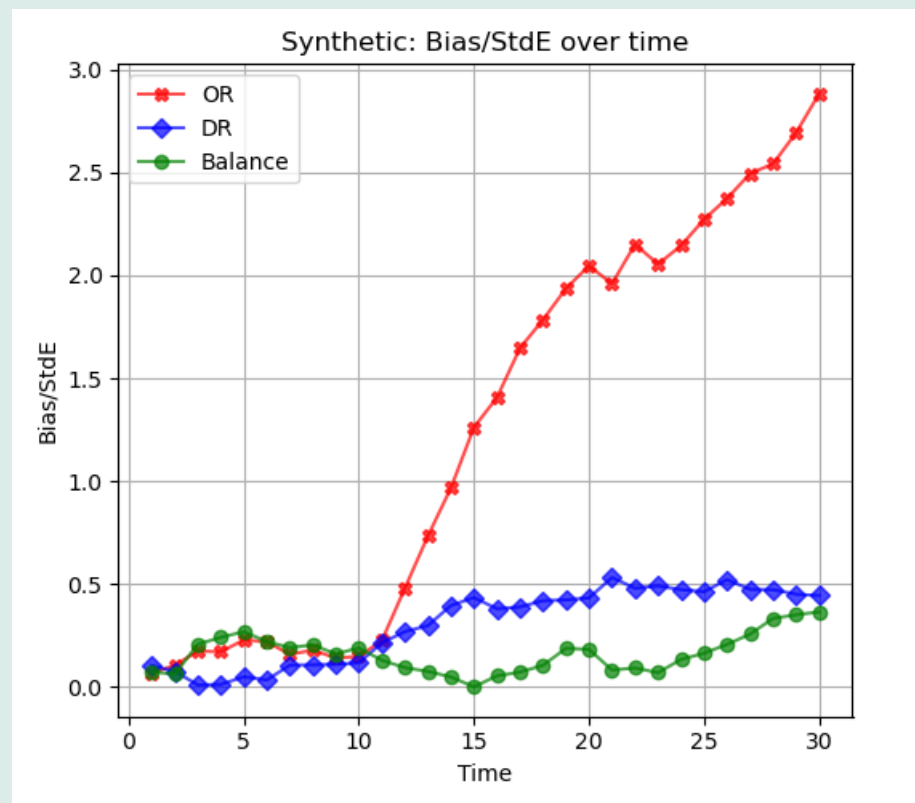


$$\frac{\text{bias}}{\text{se}} = 1 \implies \text{coverage} = 83\%$$



$$\frac{\text{bias}}{\text{se}} = 2 \implies \text{coverage} = 48\%$$

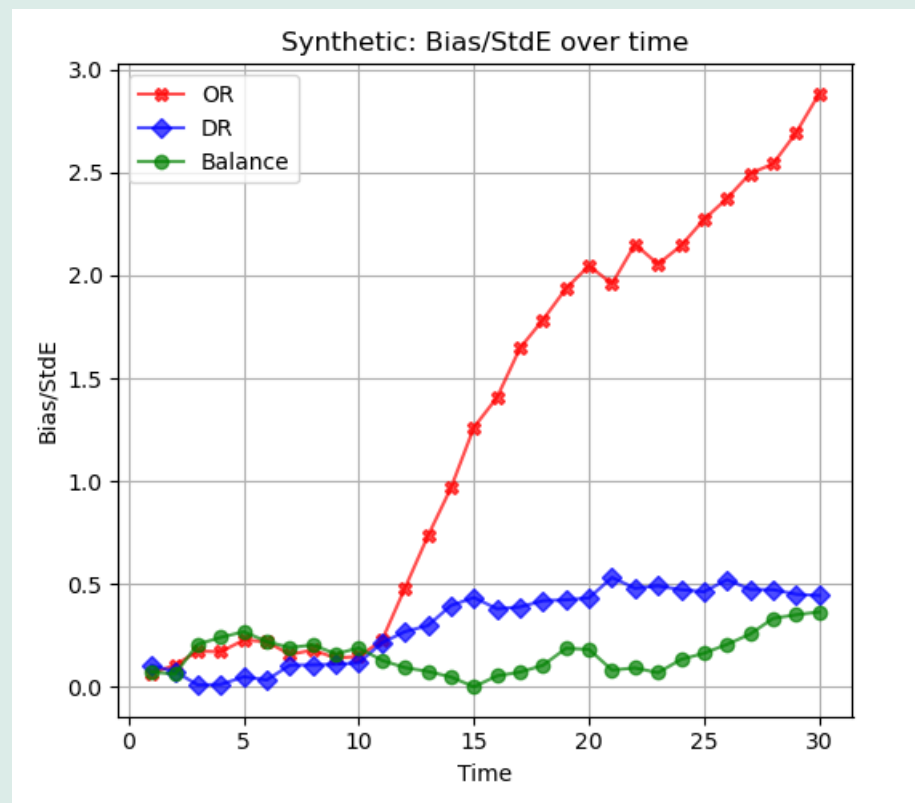
People Report This



From *Stable Estimation of Survival Causal Effects*. Pham et al. [arXiv:2310.02278](https://arxiv.org/abs/2310.02278)

- This plot shows bias/se for 3 estimators of the effect of a treatment on the probability of survival.
- In particular, they're estimators of the effect on survival up to time t . t is on the x -axis.
- **Q.** For small t , all 3 estimators are doing ok. When does the red one's coverage drop below 92%?
- **Q.** What can we say about the coverage of the blue one and the green one?

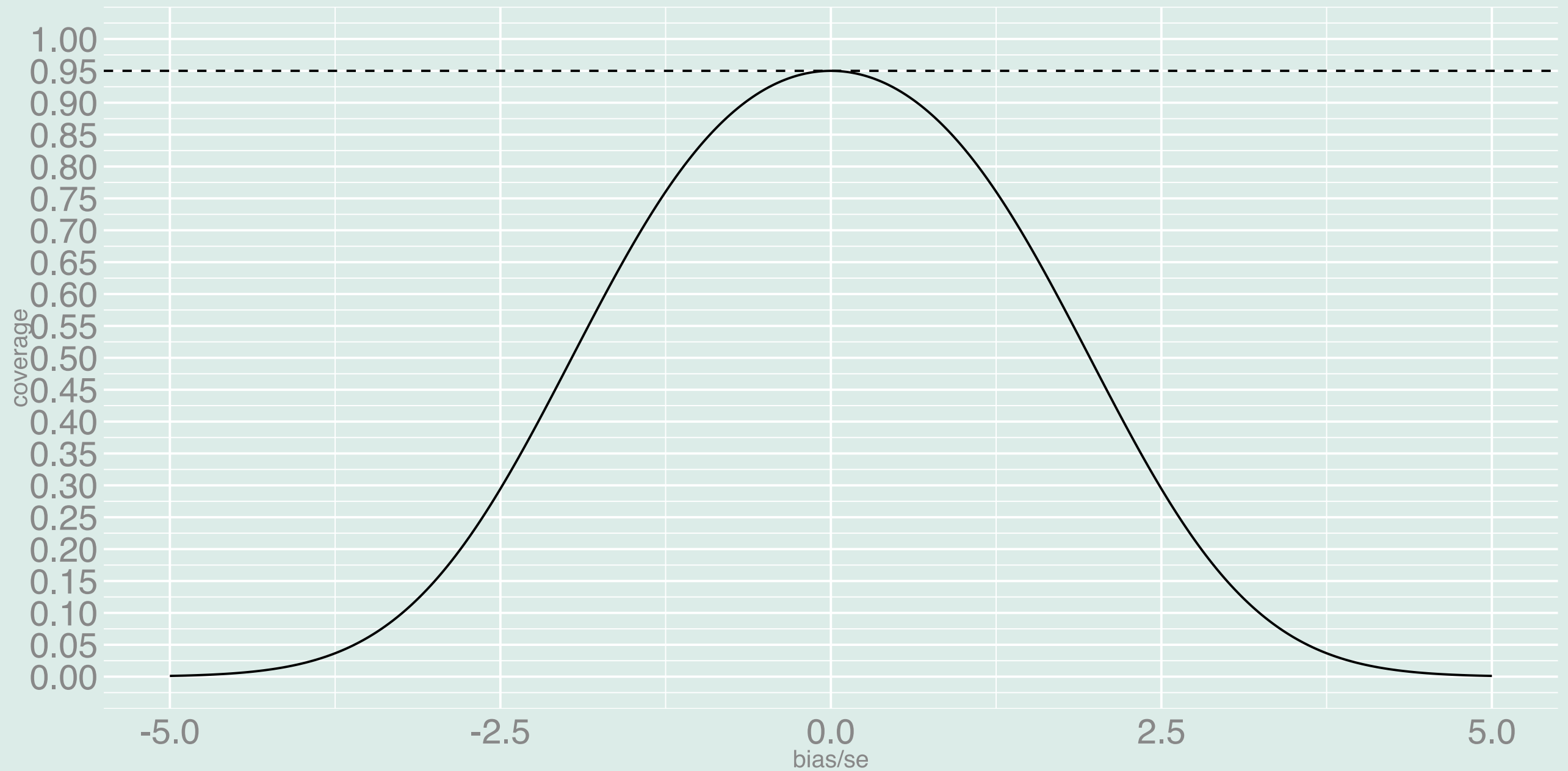
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- **Q.** For small t , all 3 estimators are doing ok. When does the red one's coverage drop below 92%?
- **Q.** What can we say about the coverage of the blue one and the green one?
- **A.** The red one's coverage drops below 92% after $t = 12$. That's where bias/se starts to exceed 1/2.
- **A.** Blue and green have bias/se close to or below 1/2 for all times t . So their coverage is roughly 92% or higher for all times t .

The Whole Bias/SE vs. Coverage Curve



A Back-of-the-Envelope Calculation

- Suppose we've got misspecification, but only a little bit.
- We're doing a pilot study with sample size $n = 100$.
- And we're sure that our estimator's bias is less than half its standard error.
- **Question 1.** What's the coverage of our 95% confidence intervals?
- Now suppose we're doing a follow-up study with sample size $n = 1600$.
- And we're using the same estimator. Bias is the same.
- **Question 2.** What's the coverage of our 95% confidence intervals?
- Assume our estimator's standard error is proportional to $\sqrt{1/n}$.
 - They almost always are. For estimators we'll talk about in this class, always.
 - That means it'll be $\sqrt{1/1600} / \sqrt{1/100} = \sqrt{1/16} = 1/4$ of what it was in the pilot study.

A Back-of-the-Envelope Calculation

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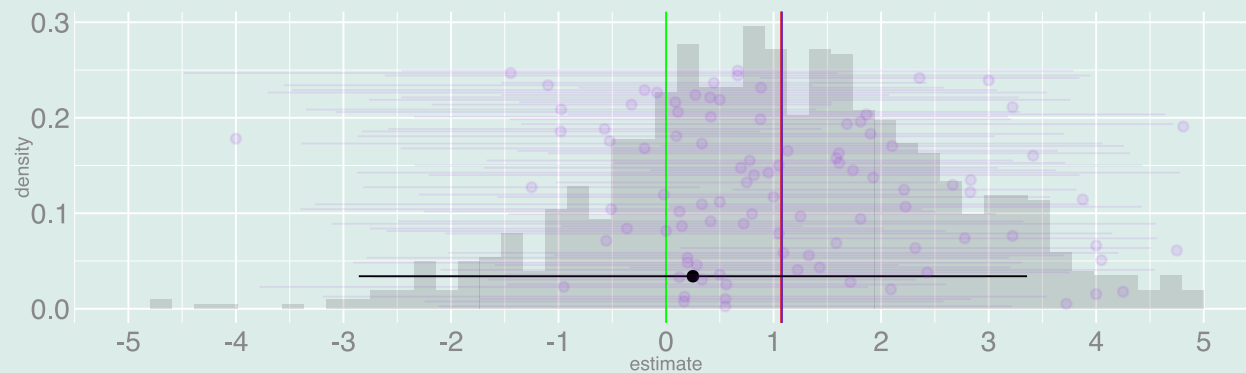
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 - They almost always are. For estimators we'll talk about in this class, always.
 - That means it'll be $\sqrt{1/1600} / \sqrt{1/100} = \sqrt{1/16} = 1/4$ of what it was in the pilot study.
- **A.** At least 48%. Here's why.

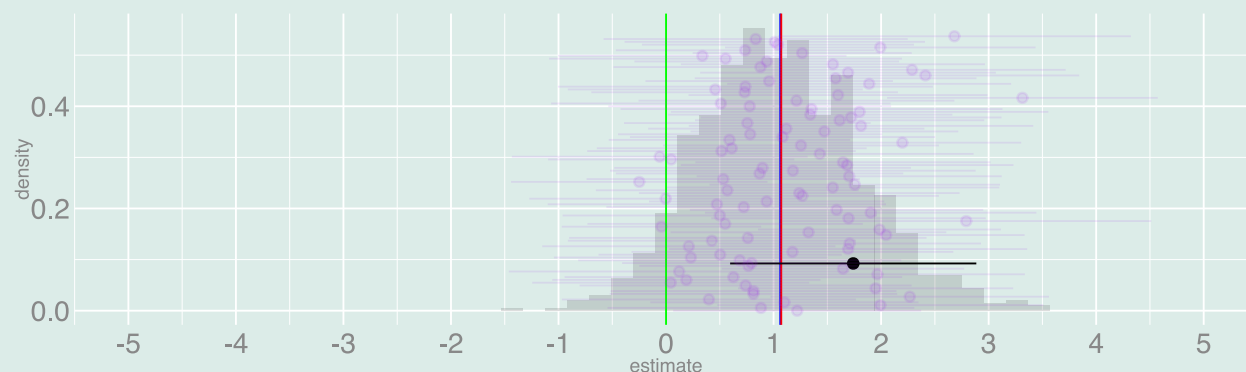
$$\text{If } \frac{\text{bias}}{\text{pilot se}} \leq 1/2 \text{ and follow-up se} = \frac{\text{pilot se}}{4}, \text{ then } \frac{\text{bias}}{\text{follow-up se}} = 4 \times \frac{\text{bias}}{\text{pilot se}} \leq 2.$$

Lessons

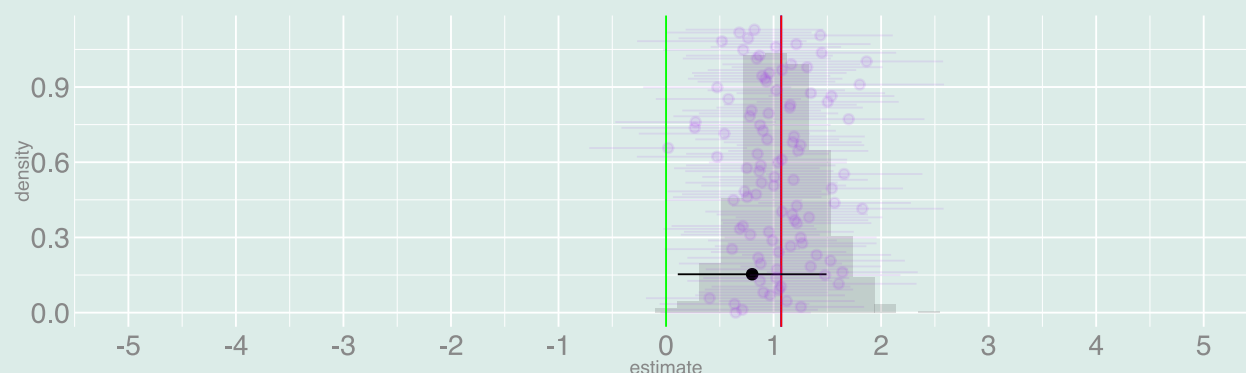
Be Careful Using Misspecified Models



sampling distribution for the line-based estimate of gym subsidy effect $\tau(100)$ at sample sizes $n=10$



sampling distribution for the line-based estimate of gym subsidy effect $\tau(100)$ at sample sizes $n=40$

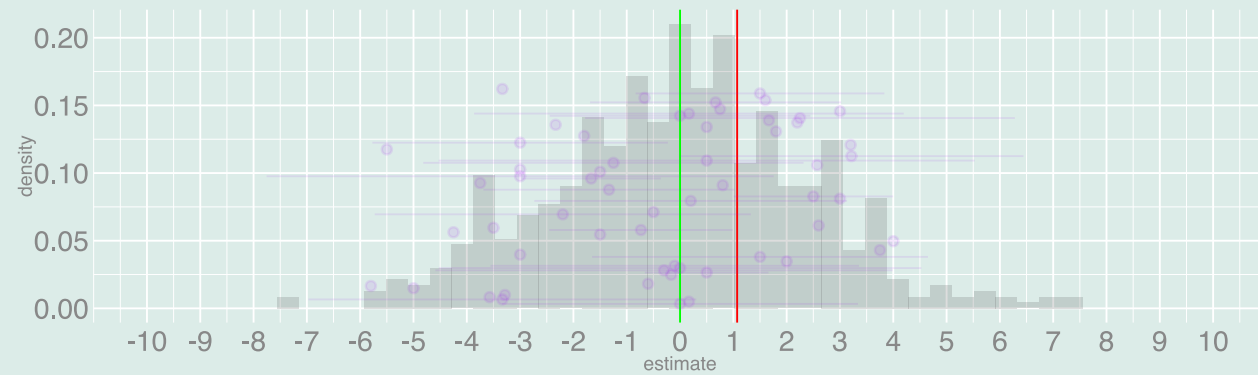


sampling distribution for the line-based estimate of gym subsidy effect $\tau(100)$ at sample sizes $n=160$

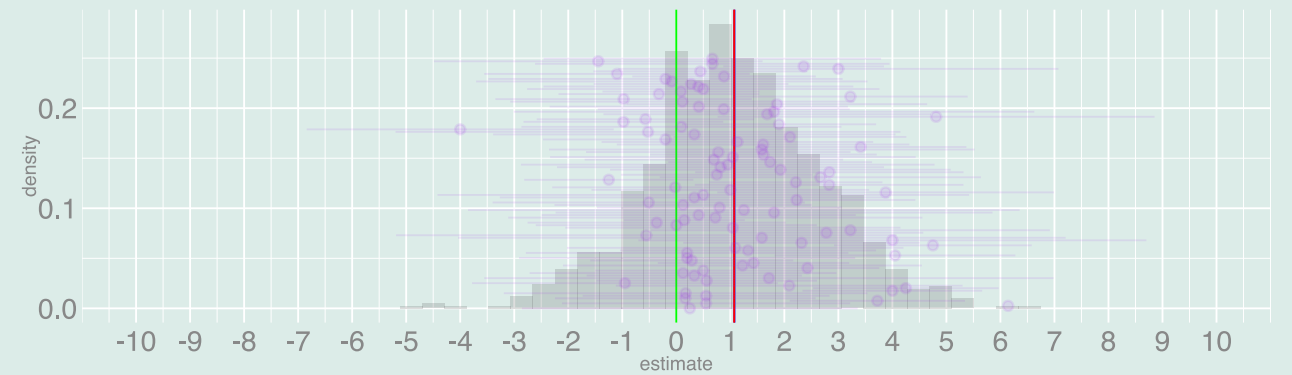
- You might have an unproblematic level of bias in a small study.
- That same amount of bias can be a big problem in a larger one.
- In our gym-subsidy example, a line was pretty badly misspecified.
- That means we have to go pretty small to get an unproblematic level of bias.
- Here are the coverage rates at sample sizes 10, 40, and 160.
- We include two versions.
 - An approximation *calculated* using our formula.
 - The *actual* coverage based on simulation.
- These differ a bit when our simplifying assumptions aren't roughly true.
 - e.g. accuracy of the normal approximation to the sampling distribution.
 - This is a bigger problem in small samples than in large ones.

n	bias	se	$\frac{\text{bias}}{\text{se}}$	calculated coverage	actual coverage
10	1.07	1.60	0.67	90%	84%
40	1.07	0.74	1.44	70%	66%
160	1.07	0.35	3.03	14%	13%

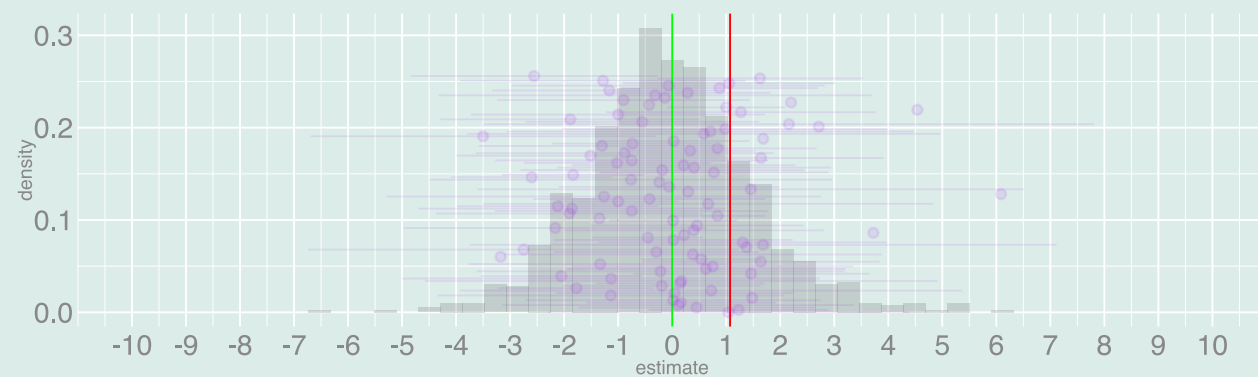
It's Best to Use a Model That's Not Misspecified



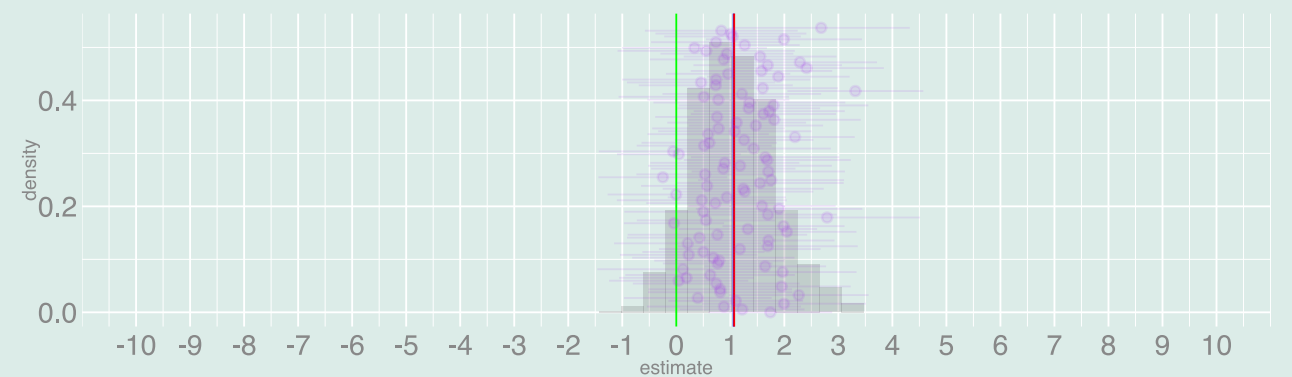
difference-in-means estimate, n=10



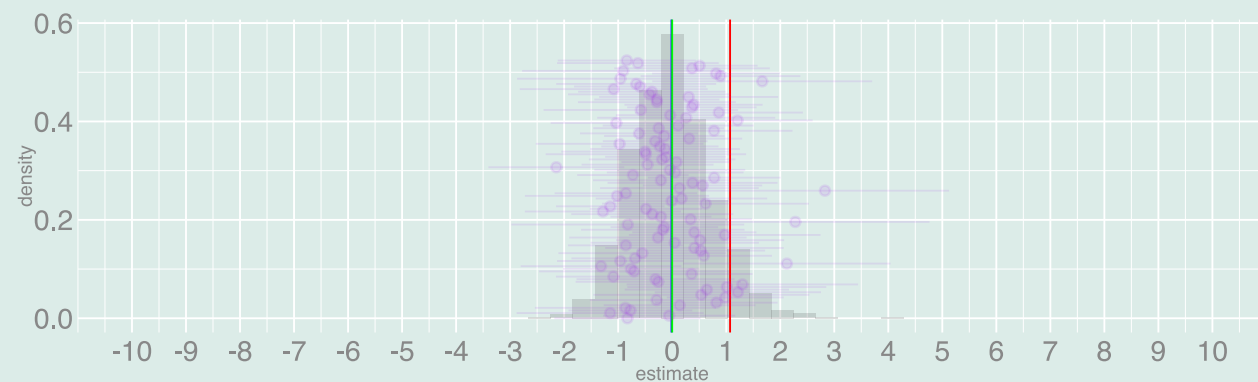
line-based estimate, n=10



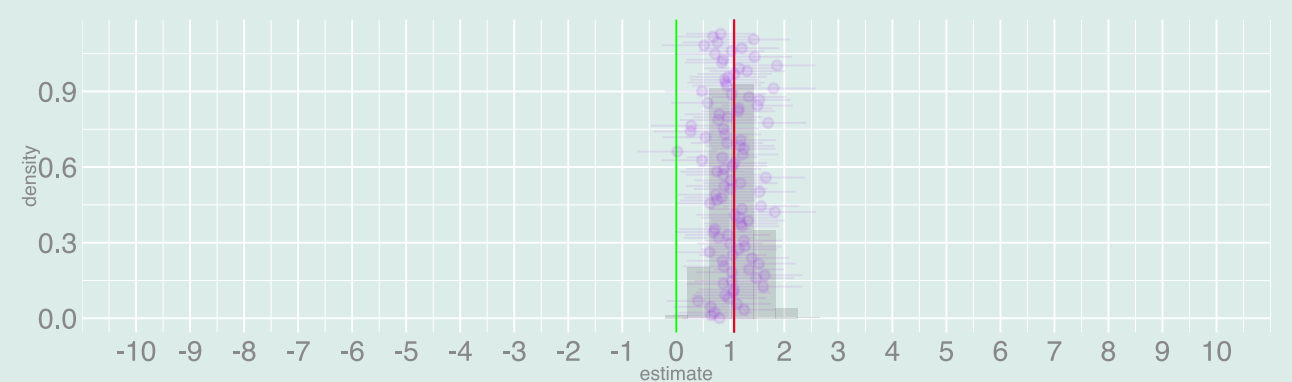
difference-in-means estimate, n=40



line-based estimate, n=40

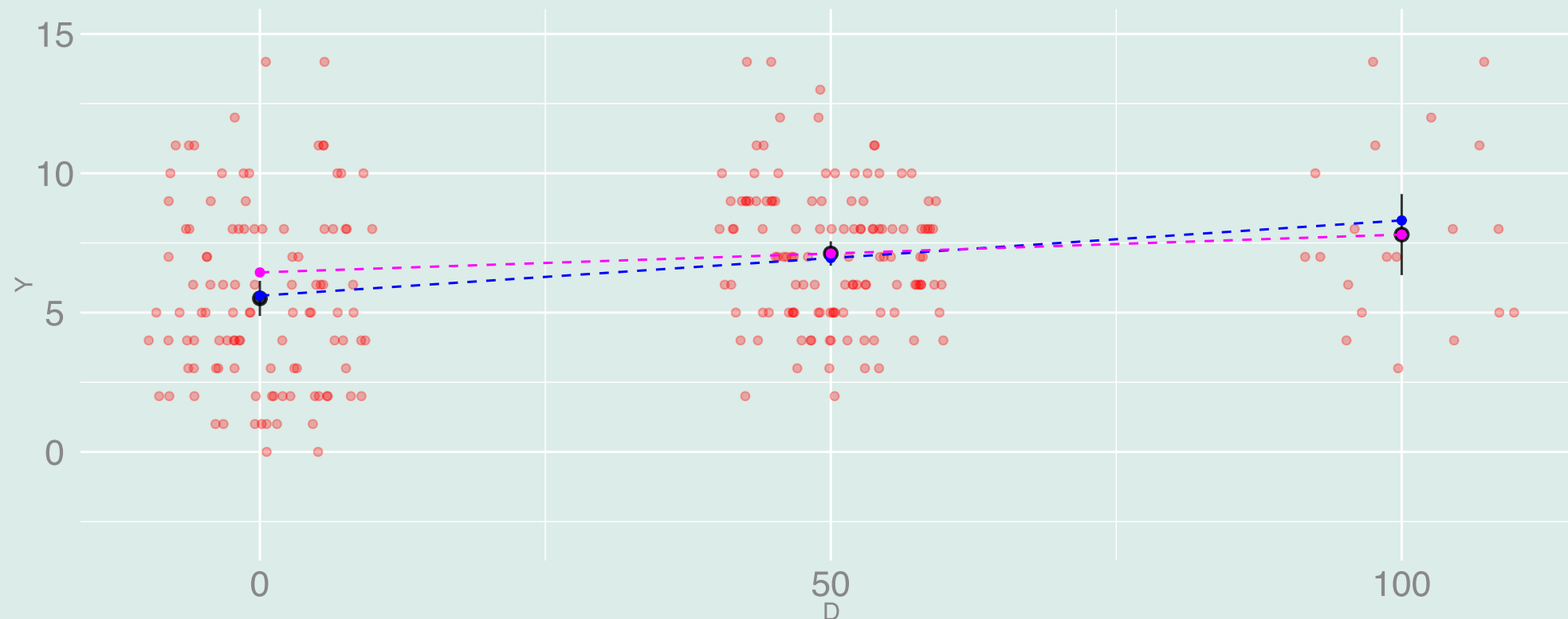


difference-in-means estimate, n=160



line-based estimate, n=160

If You're Fitting a Line, Do It Cleverly



- If we're estimating $\tau(100)$, we can use a magenta line fit to the observations with $D = 50$ and $D = 100$.
- Then, since we've effectively got a binary covariate, it goes through the subsample means.
- And we end up with the difference in means estimator. We know that's unbiased.

Later in the Semester

- We'll see a more general version of this trick.
 - We can fit misspecified models and still get unbiased estimators of summaries.
 - We just have to fit them the right way.
- We'll use *weighted least squares* with weights that depend on the summary.
- In this case, we give zero weight to observations with $D = 0$ and equal weight to the others.

